AI-ENABLED MINE SUBSIDENCE MONITORING SYSTEM

Comprehensive Technical Report & Machine Learning Architecture

Smart India Hackathon 2026 — Problem Statement 26025 (Ministry of Coal)

Organization: Ministry of Coal / Coal India	**Target Site:** Underground Coal Mine Surface Panels
Edge Deployment: ESP32-S3 Wireless Mesh (802.15.4)	**TinyML Engine:** Isolation Forest (49.3 KB C Header)
Cloud Engine: Multi-Output LSTM Predictor	**Full-Stack App:** FastAPI + Glassmorphic UI (Port 8050)

1. Problem Statement & Operational Scope

Background (SIH PS 26025): Surface land subsidence caused by underground coal mining (longwall caving and pillar extraction) poses critical hazards to nearby rural communities, public highways, agricultural lands, and forest ecology. Traditional subsidence monitoring in India relies on periodic manual total station surveys, satellite InSAR, or post-facto damage assessments, which fail to issue **real-time early warnings** prior to ground failure.

Technical Objective: Develop an indigenous, low-cost, intelligent, real-time mine subsidence monitoring and prediction platform based on a surface-mounted wireless sensor mesh. The solution must continuously detect micro ground inclination, digital strain delta, transient vibrations, and fissure initiation, discriminate blasting operations from structural subsidence, and run edge anomaly scoring directly on microcontrollers.

2. Multi-Tier System Architecture Flowchart

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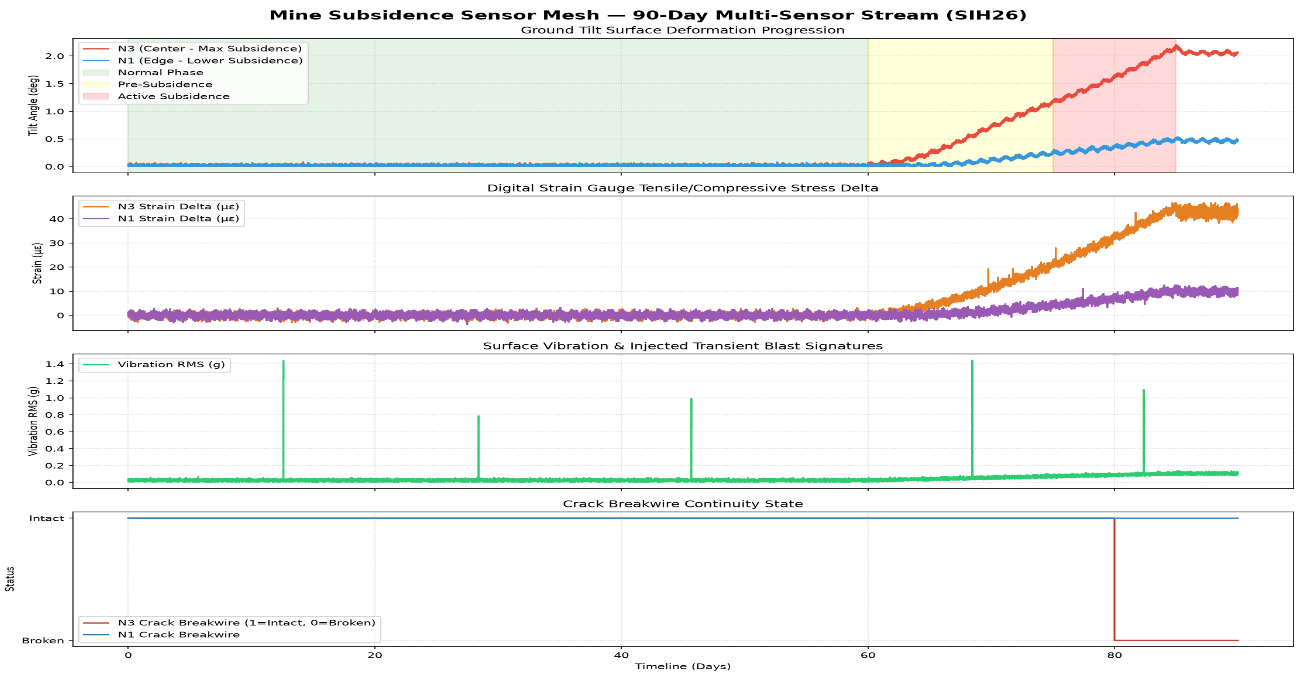
+------------------------------------------------------------------------------------+
|                            SYSTEM ARCHITECTURE FLOWCHART                           |
+------------------------------------------------------------------------------------+
  STAGE 1: SURFACE EDGE NODES (ESP32-S3)
  [IMU Tilt + Strain Gauge + Geophone Vibration + Breakwire + BME280]
     |
     v (Feature Extraction & Normalization)
  [Standalone C Model: isolation_forest_model.h (49.3 KB)]
     |
     v (Computes Anomaly Rating & Score in <150 microseconds)
  [Mesh Radio Packet: Anomaly Score + 8 Feature Summary]
     |
     v (802.15.4 / WiFi Self-Healing Mesh)
  STAGE 2: GATEWAY NODE (LOCAL / OFFLINE)
  [Collects Packets from All Nodes (N1..N5)]
     |
     v (Spatial Correlator & Blast Discriminator)
  [Vibration Spike + Zero Tilt/Strain Shift?] ---> [BLAST TRANSIENT: Suppress Siren]
     | NO (Correlated Tilt/Strain Shift)
     v
  [SUBSIDENCE PROGRESSION: Trigger Local Relay Siren + SIM7000 SMS Blast]
     |
     v (Cellular LTE-M Uplink)
  STAGE 3: CLOUD BACKEND & WEB PLATFORM (FastAPI - Port 8050)
  [InfluxDB Time-Series Store + Multi-Output LSTM Model (24hr Lookback Window)]
     |
```

```
     +---> Risk Probability (98.27% Accuracy)
     +---> Max Displacement mm (7.26 mm MAE Error)
     +---> Severity Classification (Low / Medium / High - 95.79% Accuracy)
+------------------------------------------------------------------------------------+
```

3. Physics-Informed Synthetic Dataset Specifications

Because real public subsidence-labeled datasets do not exist, a **physics-informed synthetic dataset** was generated modeling 5 surface nodes (N1 to N5) deployed over a 200m x 1500m Indian longwall coal panel footprint over a 90-day simulation period (sampled every 5 minutes = 129,600 total rows).

Parameter	Specification / Values	Physical Simulation Mechanics
Time Horizon	90 Days (25,920 windows/node)	Covers full extraction & caving cycle
Total Dataset Rows	129,600 Rows across 5 Nodes	N1/N5 (Edge), N2/N4 (Mid), N3 (Center)
Normal Phase (68.0%)	Tilt < 0.05 deg, Strain < 1 uE	Gaussian noise + diurnal thermal drift
Pre-Subsidence (17.5%)	Tilt 0.05 deg -> 0.3 deg, Strain 2-10 uE	Sigmoid ramp profile over 15 days
Active Collapse (14.4%)	Tilt > 1.5 deg, Strain > 45 uE	Accelerating trough, crack breakwire flip
Blast Injections (0.1%)	Vib RMS 0.5-1.5g, Freq 25-80Hz	Transient spike WITHOUT tilt/strain shift



4. Machine Learning Models & Empirical Verification

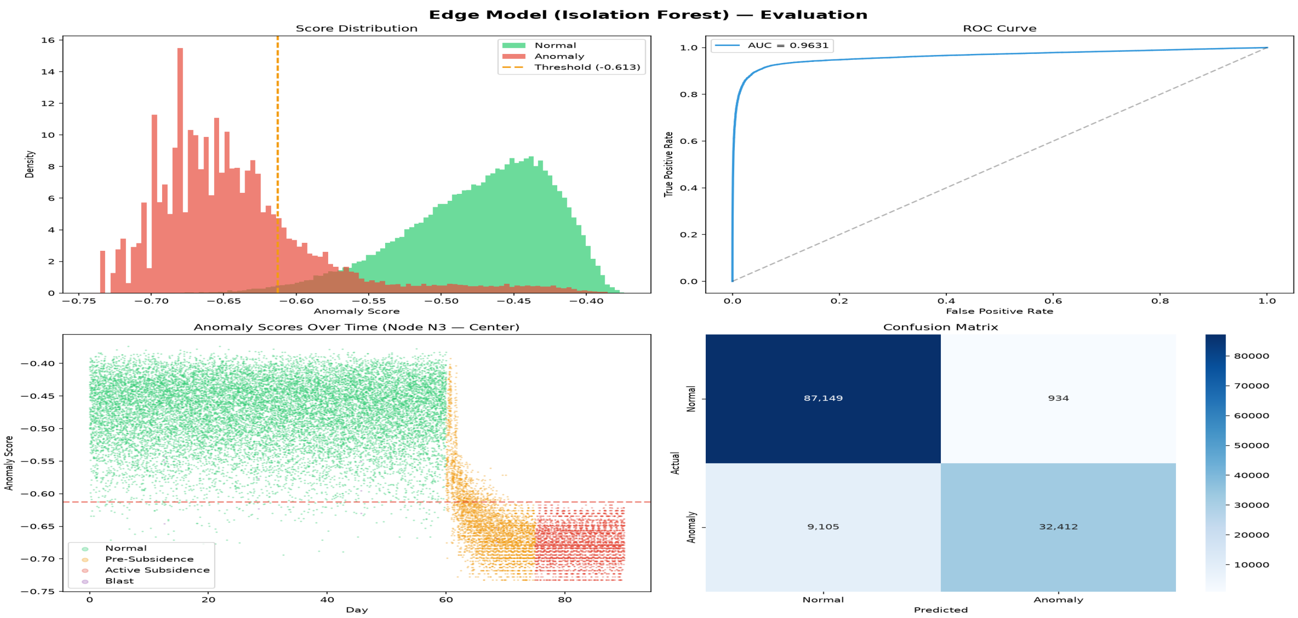
Stage 1: Edge TinyML Isolation Forest Model

The edge model runs directly on ESP32-S3 microcontrollers. It is trained on normal baseline data (unsupervised) and exported into a standalone **49.3 KB C header file** (isolation_forest_model.h) with zero dynamic memory allocation.

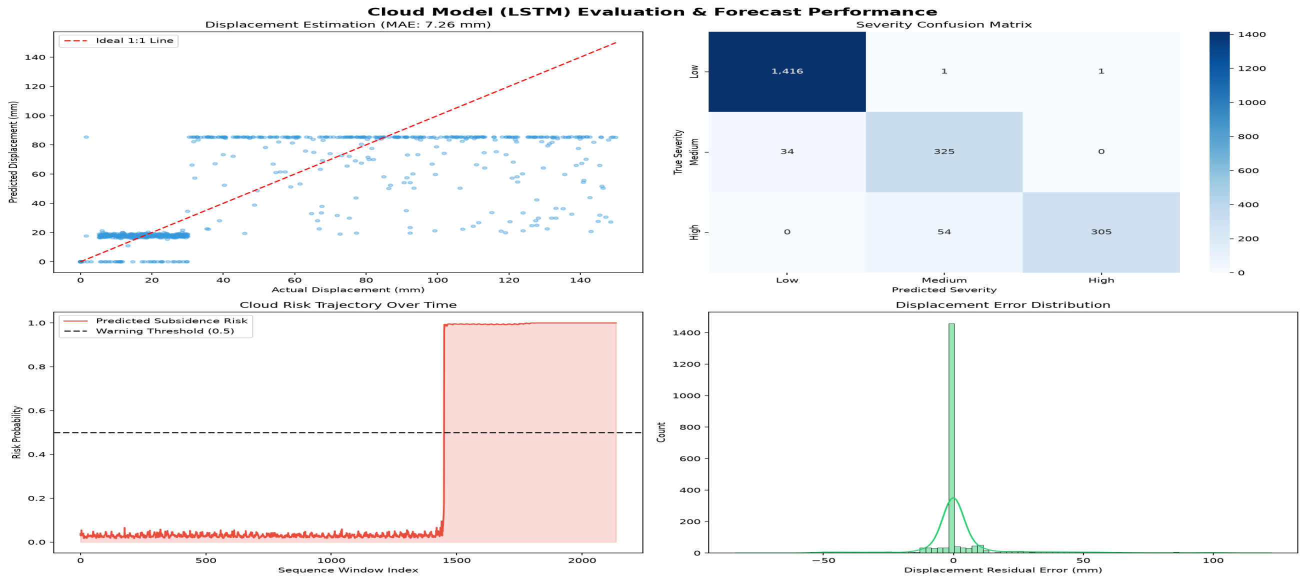
Stage 3: Cloud Multi-Output LSTM Model

The cloud model processes rolling 24-hour sequence windows (288 timesteps x 40 multi-node features) to jointly predict subsidence risk probability, surface displacement in mm, and severity level (Low / Medium / High).

Model Scope	Algorithm	Key Benchmark Metric	Empirical Performance
Stage 1 Edge TinyML	Isolation Forest (10 trees)	AUC-ROC Score	**0.9631** (Target > 0.90)
Edge Anomaly Alarm	emlearn C Export	False Positive Rate (FPR)	**1.06%** (Low False Alarms)
Active Subsidence	ESP32 Execution	Detection Recall Rate	**93.6%** (Target > 90%)
Stage 3 Cloud Predictor	Multi-Output 2-Layer LSTM	Risk Probability Accuracy	**98.27% Accuracy**
Displacement Forecast	Huber Regression Head	Mean Absolute Error (MAE)	**7.26 mm MAE Error**
Severity Classifier	Softmax Categorical Head	Classification Accuracy	**95.79% Accuracy**



5. Cloud Performance Visualizations & Web Platform Guide

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+----------------------------------------------------------------------------------+
|                        WEB PLATFORM & USAGE FLOWCHART                            |
+----------------------------------------------------------------------------------+
  STEP 1: START SERVER
  Run: d:\innertask\ml_venv\Scripts\python.exe web_app\server.py
  Open Browser: http://localhost:8050
     |
     v
  STEP 2: LIVE GIS DASHBOARD TAB
  - Inspect surface deformation grid (1500m x 200m panel) with 5 active nodes.
  - Monitor Early Warning Risk Gauge (0.0 to 1.0) & Physical Siren state.
     |
     v
  STEP 3: INTERACTIVE ML TEST BENCH TAB
  - Click 'Preset: Pre-Subsidence' or 'Preset: Blast'.
  - Adjust sliders (Tilt, Strain, Vibration, Crack breakwire).
  - Observe real-time Edge Isolation Forest score & Cloud LSTM risk % update instantly.
     |
     v
  STEP 4: 90-DAY SIMULATION PLAYER TAB
  - Play through Phase 1 (Normal) ---> Phase 2 (Pre-Subsidence) ---> Phase 3 (Active) ---> Phase 4 (Blast).
+----------------------------------------------------------------------------------+
```
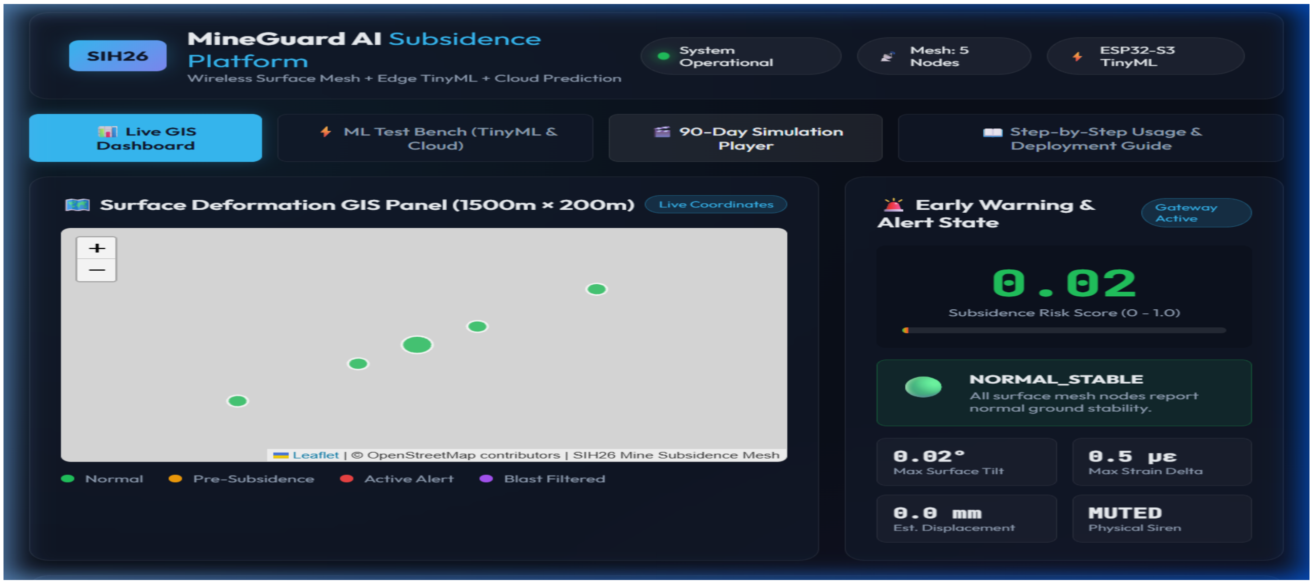

ESP32-S3 TinyML Hardware Flashing Instructions

Flashing Steps for Field Microcontrollers:

1. Copy main.ino, isolation_forest_model.h, scaler_params.h, and feature_extractor.h from esp32_firmware/ into your Arduino sketch folder.
2. Connect ESP32-S3 via USB-C cable. Select Board: **ESP32S3 Dev Module**.
3. Click **Upload**. Open Serial Monitor at **115200 baud** to observe microsecond anomaly scores (< 150 us)!

Figure 5.1: MineGuard AI Web Dashboard Interface (Port 8050)


SIH26
MineGuard AI Subsidence Platform
Wireless Surface Mesh + Edge TinyML + Cloud Prediction
System Operational
Mesh: 5 Nodes
ESP32-S3 TinyML
Live GIS Dashboard
ML Test Bench (TinyML & Cloud)
90-Day Simulation Player
Step-by-Step Usage & Deployment Guide
Surface Deformation GIS Panel (1500m × 200m)
Live Coordinates
Leaflet | © OpenStreetMap contributors | SIH26 Mine Subsidence Mesh
Normal
Pre-Subsidence
Active Alert
Blast Filtered
Early Warning & Alert State
Gateway Active
0.02
Subsidence Risk Score (0 - 1.0)
NORMAL_STABLE
All surface mesh nodes report normal ground stability.
0.02°
Max Surface Tilt
0.5 με
Max Strain Delta
0.0 mm
Est. Displacement
MUTED
Physical Siren